

Question:

A candle melts by 2 centimeters every hour that it burns. How many centimeters shorter will a candle be after burning from 1:00 PM to 5:00 PM?

Reasoning + Answer:

<think> To determine how many centimeters the candle will be shorter after burning from 1:00 PM to 5:00 PM, we need to calculate the total number of hours it burns in that time frame. The candle burns from 1:00 PM to 5:00 PM, which is a total of 4 hours. Since the candle melts by 2 centimeters every hour, we can calculate the total melting by multiplying the number of hours by the rate of melting: $4 \text{ hours} * 2 \text{ centimeters/hour} = 8 \text{ centimeters}$. </think><answer> 8 </answer>

